Two-Stage CMOS Error Amplifier for a 0.18 µm Biomedical Energy-Harvesting PMU

Conventional-inversion (non-subthreshold) LDO error amplifier • 0.18 µm CMOS (gpdk180) • Reference context: Patel, Mishra, Nagchoudhuri, "A 36 nW Power Management Unit for Solar Energy Harvesters Using 0.18 µm CMOS" (VDAT 2017)

> **Parameter classification used throughout.**
> **(A) Assumed nominal design values** — targets, overdrive choices, V_A', gate-cap estimate.
> **(B) From Paper_1 Table 1** — V_{TH}, μC_{ox} (first-order stand-ins for gpdk180).
> **(C) MUST be extracted from gpdk180** — actual V_{TH}, μC_{ox}, λ/V_A per device, pass-device gate cap. Hand values are seeds, not final.

Part 0 — Parameters Derived From Paper_1.pdf

From Table 1 of the paper (first-order 0.18µm constants). Threshold voltages and transconductance parameters are used for square-law sizing; the paper's λ and η are rejected (subthreshold / short-device values, and the whole reference amplifier is subthreshold).

Parameter	NMOS	PMOS	Use in this design
V_{th}	0.4616 V	-0.518 V ($\lvert V_{tp}\rvert$=0.518)	Used directly for V_{GS}/headroom
$k = \mu C_{ox}$	316 µA/V²	62.5 µA/V²	Used directly for W/L square-law
η (subthreshold slope)	1.4	1.6	Rejected — subthreshold only
λ (V^{-1})	0.92	-0.95	Rejected for gain (see note)

> **Note on λ.** λ≈0.9/V implies $r_o \approx 1/(\lambda \cdot I_D)$ of only a few hundred kΩ — a subthreshold/minimum-length artifact. For our strong-inversion devices with longer L, output resistance is far higher. Gain is modelled with an Early-voltage approach, $r_o = V_A/I_D$, $V_A = V_A' \cdot L$, with nominal **$V_A' \approx 8$ V/µm** (flagged (C), extract in Cadence).

Application context (reference only): PMU regulates ≈0.95 V from ≈0.98–1.33 V harvested input; error amp is a cascoded two-stage; C_L=10 µF at the LDO output; pass element is a strong-inversion PMOS 600 µm/0.18 µm; reference error amp = 11.8 nA, 46.39 dB, 206 kHz UGB (subthreshold). We deliberately diverge on inversion region, sizing and currents.

Part 1 — Final Proposed Specifications

#	Spec	Target	Why
1	Technology	0.18 µm CMOS (gpdk180)	Project constraint; matches paper
2	Supply V_{DD}	1.8 V	Nominal 0.18µm rail; harvester boost/charge-pump supplies unregulated ~1.8 V that the LDO regulates down; gives headroom for strong inversion
3	Input common-mode	0.5 V (≈V_{REF})	Low CM favors PMOS input pair
4	Output range (OUT)	0.15 – 1.6 V	Must swing to drive PMOS pass gate on/off
5	V_{REF}	0.5 V	Half of V_{OUT} with R1=R2
6	V_{FB}	0.5 V at regulation	= V_{REF} at balance
7	DC gain A_v	≥ 60 dB (aim 65–68)	Higher than paper 46 dB → better line/load regulation
8	UGB	≈ 1.3 MHz	Enough loop speed; not excessive
9	Phase margin	≥ 60° (design ~70°)	LDO loop stability with real output pole
10	Slew rate	≥ 0.5 V/µs (design ~1)	Fast enough; 10 µF output slows loop anyway

11	Load cap (amp)	C_L = 1 pF	= pass-PMOS gate cap + routing (est.)
12	Max quiescent current	≤ 5 µA (design 4)	µA-class because strong inversion
13	Power	≤ 9 µW (design ~7.2)	Realistic strong-inversion; harvester ~1 mW
14	Output swing	0.15 – 1.6 V	Adequate pass-gate drive
15	CMRR	≥ 60 dB	Reject CM noise on V_{REF} path
16	PSRR (LF)	≥ 50 dB	Noisy harvested rail; sets LDO PSRR floor
17	Input offset	≤ 5 mV (systematic ~0)	Offset → V_{OUT} error via divider
18	Temperature	-40 to +85 °C	Wearable/implant environment
19	Corners	TT/SS/FF/SF/FS	Manufacturability
20	Loop stability	PM ≥ 45° over corners w/ 10 µF+ESR	Must stay stable inside LDO

Part 2 — Architecture Selection

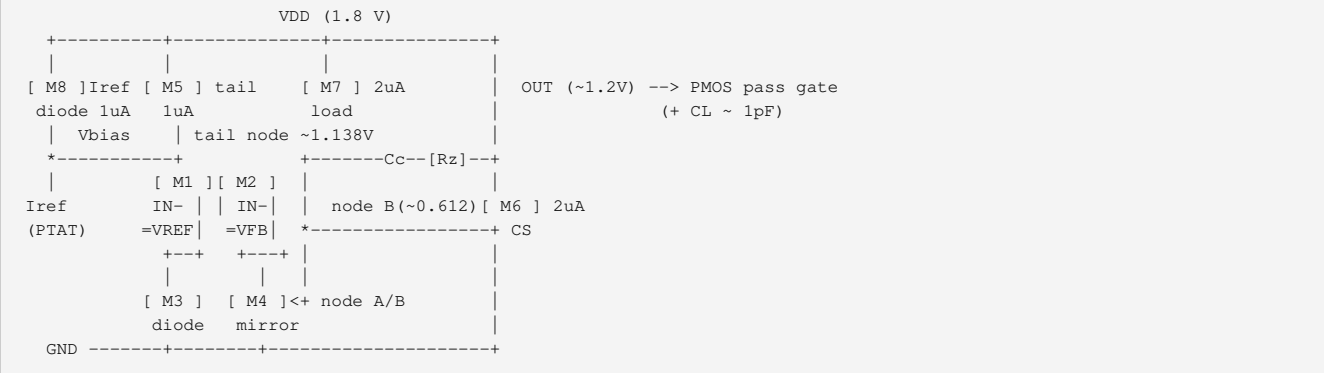
Chosen: conventional two-stage Miller op-amp — PMOS differential input pair + NMOS current-mirror load (Stage 1), NMOS common-source + PMOS current-source load (Stage 2), Miller C_c with nulling resistor R_z.

PMOS input pair, because:

- **Input CM is low (0.5 V).** NMOS pair needs $V_{ICM,min} \approx V_{ov,tail}+V_{tn}+V_{ov1} \approx 0.71$ V — our 0.5 V is below that. PMOS pair senses down toward ground.
- **LDO compatibility:** amp drives a PMOS pass gate; PMOS-input / NMOS-CS-output gives the right DC level and swing.
- **Noise:** PMOS has lower 1/f flicker noise — important for biomedical low-frequency signals.
- Trade-off accepted: PMOS gm ~5x lower than NMOS, so input devices are a bit larger — fine at these currents.

Cascoded second stage NOT used (unlike the paper). The paper cascodes to recover gain lost in subthreshold at ~1 V. With 1.8 V and strong inversion a plain two-stage reaches ≥60 dB at L≈1 µm. Cascoding would steal output headroom (bad, since OUT must swing to drive the pass gate) and add poles. A simple Miller two-stage is sufficient and superior here — the key justified divergence from Paper_1.

Part 3 — Transistor-Level Topology



Dev	Type	Function	Connection	Region	I_D
M1	PMOS	Input - (V_{REF})	S→tail, G=V_{REF}, D→node A(M3)	sat	0.5 µA
M2	PMOS	Input + (V_{FB})	S→tail, G=V_{FB}, D→node B(M4,M6 gate)	sat	0.5 µA
M3	NMOS	Mirror diode	G=D=node A, S=GND	sat	0.5 µA
M4	NMOS	Mirror out (active load)	G=node A, D=node B, S=GND	sat	0.5 µA
M5	PMOS	Tail current source	S=VDD, G=V_{bias}, D=tail node	sat	1 µA

M6	NMOS	2nd-stage gain (CS)	G=node B, S=GND, D=OUT	sat	2 µA
M7	PMOS	2nd-stage load (I-source)	S=VDD, G=V_{bias}, D=OUT	sat	2 µA
M8	PMOS	Bias mirror diode	S=VDD, G=D=V_{bias}, I_{ref} in	sat	1 µA
Cc	cap	Miller comp	node B ↔ OUT	—	—
Rz	res	Nulling (zero)	series with Cc	—	—

I_{ref} (1 µA) comes from the PMU PTAT reference. M8 sets V_{bias}; M5 mirrors x1 (1 µA), M7 mirrors x2 (2 µA).

Part 4 — Bias Conditions and Region Verification

Design overdrives (A): V_{ov1}=0.12 V (input); V_{ov}(NMOS mirror & M6)=0.15 V; V_{ov}(PMOS sources M5/M7/M8)=0.20 V.

Node DC (V_{DD}=1.8, V_{ICM}=0.5): tail node = 0.5+(0.518+0.12)=**1.138 V**; node A/B = V_{GS}(M3)=0.462+0.15=**0.612 V**; OUT ≈ **1.2 V** (set by loop); V_{bias}=1.8-(0.518+0.20)=**1.082 V**.

Dev	V_{GS}/V_{SG}	vs V_{TH}	V_{DS}/V_{SD}	vs V_{OV}	Region
M1/M2 (P)	0.638	>0.518 OK	0.526	>0.12 OK	sat, strong
M3 (N)	0.612	>0.462 OK	0.612	>0.15 OK	sat
M4 (N)	0.612	>0.462 OK	0.612	>0.15 OK	sat
M5 (P)	0.718	>0.518 OK	0.662	>0.20 OK	sat
M6 (N)	0.612	>0.462 OK	1.2	>0.15 OK	sat
M7 (P)	0.718	>0.518 OK	0.6	>0.20 OK	sat
M8 (P)	0.718	>0.518 OK	0.718	>0.20 OK	sat (diode)

Every device: $V_{GS}>V_{TH}$ and $V_{DS}\geq V_{OV}$ → all in saturation, none in subthreshold. Constraint satisfied.

(C) Extract from gpdk180: real V_{TH} (body effect raises $|V_{tp}|$ of M1/M2), μC_{ox} at these V_{ov}, and λ per device.

Part 5 — W/L for Every Transistor

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W/L = 2·I_D / (µC_ox · V_ov²) — first-order seed only; final values optimized on gpdk180.
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- **M1,M2** (P, 0.5µA, 0.12): 2·0.5µ/(62.5µ·0.0144)=1µ/0.90µ = **1.11**
- **M3,M4** (N, 0.5µA, 0.15): 1µ/(316µ·0.0225)=1µ/7.11µ = **0.14**
- **M5** (P, 1µA, 0.20): 2µ/(62.5µ·0.04)=2µ/2.5µ = **0.80**
- **M6** (N, 2µA, 0.15): 4µ/(316µ·0.0225)=4µ/7.11µ = **0.56**
- **M7** (P, 2µA, 0.20): 4µ/(62.5µ·0.04) = **1.60**
- **M8** (P, 1µA, 0.20): = **0.80** (=M5, so M7 = 2x)

Zero-systematic-offset check: need $(W/L)_6/(W/L)_4 = 2\cdot(W/L)_7/(W/L)_5$ → 0.56/0.14 = 4.0 and 2·1.6/0.8 = 4.0. Matches → V_{GS6} equals node-B voltage automatically, systematic offset ~0.

Channel-length choices (not all minimum): input M1/M2 L=1µm (low flicker noise, better matching, higher r_o); mirror M3/M4 L=2µm (tight matching + high r_o); tail M5, load M7 L=1µm (high r_o for PSRR/gain); M6 L=1µm (gain vs non-dominant pole balance). Nothing at min-L: min-L kills r_o/gain, worsens λ, matching and 1/f noise, and we are not bandwidth-starved.

Trade-offs: higher W/L → higher gm (gain, UGB) but higher gate cap (lower f_{p2}, worse PM) and area; higher L → higher r_o/gain, matching, lower noise, but higher cap and lower f_t; higher I_D → higher gm and SR but more power. Improve matching by increasing area at constant W/L (common-centroid), not by re-sizing.

Part 6 — First Stage

- $g_{m1} = 2\cdot I_{D1}/V_{ov1} = 2\cdot 0.5µ/0.12$ = **8.33 µS**
- Tail current $I_5 = 2\cdot I_{D1}$ = 1 µA
- r_{o2} (P,L=1) = V_A/I_D = 8/0.5µ = 16 MΩ; r_{o4} (N,L=2) = 16/0.5µ = 32 MΩ
- $R_{out1} = r_{o2} \parallel r_{o4}$ = 16M || 32M ≈ **10.7 MΩ**

- $A_{v1} = g_{m1} \cdot R_{out1} = 8.33\mu \cdot 10.7M \approx$ **89 → 39 dB**

Internal pole at node B ($C_B \approx 0.3$ pF): f ≈ 50 kHz — becomes dominant via C_c (Part 8). Assumptions: square-law gm; r_o from assumed V_A; gain accurate to ~±40% until λ extracted, hence the ≥60 dB target keeps margin.

Part 7 — Second Stage

- $I_{D6} = 2$ µA; $g_{m6} = 2 \cdot 2\mu/0.15 =$ **26.7 µS**
- r_{o6} (N,L=1)=8/2µ=4 MΩ; r_{o7} (P,L=1)=4 MΩ → $R_{out2} =$ **2 MΩ**
- $A_{v2} = g_{m6} \cdot R_{out2} = 26.7\mu \cdot 2M =$ **53.3 → 34.5 dB**
- Output swing: $V_{ds6,sat}$(0.15) up to V_{DD}-$V_{sd7,sat}$(1.6) → 0.15–1.6 V

Overall gain $A_v = A_{v1} \cdot A_{v2} = 89 \cdot 53.3 \approx$ **4740 → 73.5 dB** (nominal $V_A' = 8$). If actual $V_A' \approx 5$, A_v drops to ~62 dB — still ≥60 dB. If Cadence shows <60 dB, increase L of M4/M6/M7 to 1.5–2 µm before changing currents.

Part 8 — Miller Compensation

Choose **C_c = 1 pF** (with C_L=1 pF; satisfies $C_c \geq 2.2 \cdot (g_{m1}/g_{m6}) \cdot C_L = 0.69$ pF).

- **UGB** $= g_{m1}/(2\pi C_c) = 8.33\mu/(2\pi \cdot 1p) =$ **1.33 MHz**
- **Dominant pole** $f_{p1} = 1/(2\pi \cdot g_{m6} \cdot R_{out2} \cdot R_{out1} \cdot C_c) \approx$ **278 Hz**
- **Non-dominant pole** $f_{p2} = g_{m6}/(2\pi C_L) = 26.7\mu/(2\pi \cdot 1p) =$ **4.25 MHz** (=3.2×UGB)
- **RHP zero** $z = g_{m6}/(2\pi C_c) = 4.25$ MHz — cancel with nulling resistor
- **R_z** $= 1/g_{m6} = 37.5$ kΩ; use **$R_z \approx$ 40–60 kΩ** (60 kΩ places a mild LHP zero near f_{p2} to boost PM). Implement as triode PMOS tracking $1/g_{m6}$.
- **Phase margin** ≈ 90° - atan(UGB/f_{p2}) = 90° - 17.4° = **~72°** (zero cancelled)

Part 9 — Slew Rate

- C_c-limited: **SR = I_5/C_c = 1 µA/1 pF = 1.0 V/µs** (≥0.5 target)
- Output stage can source/sink ~2 µA/1 pF = 2 V/µs → input stage is the limiter, SR = 1 V/µs.

Trade-off: SR ∝ I_5/C_c and UGB = g_{m1}/C_c. Raising SR at fixed C_c needs more I_5 (power) or higher V_{ov1} (lower gm/UGB). No conflict here — the 10 µF LDO output makes the large-signal loop response far slower than 1 V/µs, so we do not spend power chasing higher SR.

Part 10 — Power

Branch	Current
Bias diode M8 (I_{ref})	1.0 µA
Tail M5 (Stage 1)	1.0 µA
Stage 2 (M6/M7)	2.0 µA
Total I_Q	**4.0 µA**

P = $V_{DD} \cdot I_Q$ = 1.8 V · 4.0 µA = 7.2 µW. Strong inversion makes µA-class unavoidable (vs the paper's 11.8 nA subthreshold). At harvested power up to ~1 mW, 7.2 µW is <1% overhead. We deliberately do NOT chase 36 nW (a subthreshold-only figure). If power must drop: halve Stage-2 to 1 µA and I_{ref} to 0.5 µA → ~2 µA / 3.6 µW, at some cost in PM and gain.

Part 11 — LDO-Specific Requirements

- V_{REF} → M1 gate (inverting sense); V_{FB} → M2 gate, $V_{FB} = V_{OUT} \cdot R2/(R1+R2)$.
- OUT → PMOS pass gate: if V_{OUT} dips, $V_{FB} < V_{REF}$, loop drives OUT down, increases pass V_{SG}, delivers more current (negative feedback).
- Line regulation ∝ 1/loop-gain; high A_v keeps rail variation off V_{OUT}.
- Load regulation ∝ $1/(g_{m,pass} \cdot A_v \cdot \beta)$; higher amp gain → smaller ΔV_{OUT} vs load step.
- Bandwidth → transient recovery; ~1.3 MHz ample given the 10 µF pole.
- Phase margin → loop stability; amp PM ~72° so total loop PM stays ≥45°.

- Output cap+ESR: 10 µF sets dominant loop pole; ESR zero (10 mΩ–1 Ω) can aid/hurt PM. Separate from amp compensation.
- Excessive amp bandwidth undesirable: pole interaction collapses PM, couples HF noise, wastes power — hence UGB capped ~1.3 MHz.
- Amp's own load cap = pass-gate cap ≈ 1 pF, NOT the 10 µF output cap.
- LDO values (biomedical): V_{OUT}=1.0 V, V_{REF}=0.5 V, R1=R2.

Part 12 — Biomedical Application Justification

- Low supply (1.8 V rail, 1.0 V output): compatible with wearable/implant CMOS front-ends and safe logic.
- Low power (7.2 µW): leaves most harvested energy for sensor/radio; long battery-free operation.
- Small area: two-stage + small caps; no bulky cascodes.
- Low noise: PMOS input + long L minimizes 1/f noise near physiological bands (ECG/EEG/EMG, sub-kHz).
- Adequate bandwidth (1.3 MHz): fast enough for duty-cycled load steps.
- Stable regulation (PM ~72°, ≥60 dB): clean rail for sensitive readout.
- Temperature (-40..85 °C): verified by corner/temp sweeps; PTAT bias tracks temperature.
- Battery-free/harvesting: µW budget aligns with solar/thermal/RF harvesters.

> **Disclaimer:** This is a circuit-level design targeted toward biomedical/healthcare electronics. It is NOT claimed to be medically safe or suitable for human implantation; that requires separate qualification.

Part 13 — Cadence Virtuoso Implementation

1. Create library (errAmp_lib), technology = gpdk180.
2. Attach tech lib (gpdk180) so device models/rules load.
3. New schematic cellview err_amp.
4. Place devices: pmos1v for M1,M2,M5,M7,M8; nmos1v for M3,M4,M6.
5. Set W/L per Part 16 table (use m/multiplier for arrays).
6. Bias: idc 1 µA into M8 diode (or PTAT sub-block).
7. Add V_{DD} = vdc 1.8 V.
8. Inputs: V_{REF} vdc 0.5 V on M1; on M2 vdc 0.5 V + vsin/vpulse for test.
9. Add C_c 1 pF + R_z 40–60 kΩ between node B and OUT.
10. DC op-point; save DC operating point.
11. Check regions: confirm each device region=sat, read id/vgs/vds/vth/gm/gds/vdsat.
12. AC analysis: ac mag=1 on diff input, 1 Hz–100 MHz.
13. DC gain: dB20(VF("/OUT")) at low freq.
14. UGB: cross(dB20(VF("/OUT")),0,1).
15. Phase margin: phaseMargin() at UGB.
16. Transient: step V_{FB}; tran to 50 µs.
17. Slew rate: slewRate() on OUT edges.
18. Power: -V_{DD}·I(V_{DD}).
19. Corner analysis: ADE-XL, gpdk180 TT/SS/FF/SF/FS.
20. Temperature: sweep -40/27/85 °C.

(C) Take from gpdk180 not hand calc: actual V_{TH} (with body effect), μC_{ox}/vdsat, gm, gds→λ, and the pass-device C_{gs}/C_{gd} to fix real C_L. Re-tune C_c/R_z to measured g_{m1}, g_{m6}.

Part 14 — Testbench

Element	Value
V_{DD}	1.8 V dc
V_{IN+} (M2 gate)	0.5 V dc + AC mag (diff)

V_{IN-} (M1 gate)	0.5 V dc + AC mag (opposite phase)
Common-mode	0.5 V
AC magnitude	1 V total differential
DC bias	I_{ref} = 1 µA
C_L	1 pF (models pass gate)
C_c / R_z	1 pF / 40–60 kΩ
Output load	1 pF (+ optional 1 MΩ)

Measurements: DC gain = 20·log10(V_{out}/V_{in}) at low f; UGB = f where gain=0 dB; PM = 180° + phase(V_{out}) at UGB; SR = max dV_{out}/dt on large step (unity buffer); Power = V_{DD}·I_{supply}; Offset = dc source in series with one input swept until V_{OUT}=ideal DC. Also run CMRR (CM ac) and PSRR (ac on V_{DD}).

Part 15 — Corner and Robustness Analysis

Runs: {TT, SS, FF, SF, FS} × {-40, 27, 85 °C} (15 runs, ADE-XL). Check at every corner: DC gain, UGB, phase margin, slew rate, power, V_{ICM} range, output swing, and that all devices stay in saturation. Worst cases: SS+85°C lowers gm/gain/UGB; FF-40°C can drop PM and raise power; SF/FS stress input-pair and mirror matching (offset). Pass: gain ≥55 dB, PM ≥45°, all devices in saturation.

Part 16 — Final Consolidated Tables

Performance

Parameter	Target	Calculated	Reason
Supply	1.8 V	1.8 V	Headroom for strong inversion
DC gain	≥60 dB	~73 dB (62 dB worst V_A)	Line/load regulation
UGB	~1.3 MHz	1.33 MHz	Loop speed
Phase margin	≥60°	~72°	LDO stability
Slew rate	≥0.5 V/µs	1.0 V/µs	Transient
I_Q	≤5 µA	4.0 µA	Low power
Power	≤9 µW	7.2 µW	Harvester budget
C_L	1 pF	1 pF	Pass gate
C_c / R_z	—	1 pF / ~50 kΩ	Compensation
Offset (systematic)	~0	~0 by ratio	Matched Stage 2

Device Sizing (seeds for gpdk180)

MOS	Type	W	L	W/L	I_D	V_{OV}	Function
M1	P	1.1 µm	1 µm	1.11	0.5 µA	0.12	Input - (V_{REF})
M2	P	1.1 µm	1 µm	1.11	0.5 µA	0.12	Input + (V_{FB})
M3	N	0.28 µm	2 µm	0.14	0.5 µA	0.15	Mirror diode
M4	N	0.28 µm	2 µm	0.14	0.5 µA	0.15	Mirror load
M5	P	0.80 µm	1 µm	0.80	1.0 µA	0.20	Tail source
M6	N	0.56 µm	1 µm	0.56	2.0 µA	0.15	2nd-stage CS
M7	P	1.60 µm	1 µm	1.60	2.0 µA	0.20	2nd-stage load

M8	P	0.80 µm	1 µm	0.80	1.0 µA	0.20	Bias diode

Widths <0.24 µm min: bump W to min and scale L to hold W/L, e.g. M3/M4 → W=0.30 µm, L=2.1 µm.

Part 17 — Iterative Design Procedure

Flow: hand calc → seed W/L → DC op-point → extract $I_D/V_{GS}/V_{DS}/V_{OV}$/gm/gds → AC (gain/UGB/PM) → adjust → transient (SR/settling) → corners → freeze.

Symptom	Change
Gain too low	Increase L of M4/M6/M7 (higher r_o); last resort cascode M4/M7
UGB too low	Increase g_{m1} (more I_{D1} or lower V_{ov1}) or reduce C_c
PM too low	Increase C_c, or g_{m6}; tune R_z for LHP zero at f_{p2}; cut C_L parasitics
SR too low	Increase I_5 or reduce C_c
Power too high	Reduce I_5, I_{D6}, I_{ref} (accept lower UGB/SR)
Output won't swing	Lower V_{ov} of M6/M7; verify OUT DC set by loop
Device leaves saturation	Check V_{DS} vs V_{dsat}; add headroom or re-bias
V_{ICM} insufficient	Lower V_{ov1} and/or V_{ov5}; confirm PMOS pair senses 0.5 V

Do not treat seed W/L as final — real gpdk180 V_{TH} (body effect), $\mu C_{ox}(V_{ov})$ and λ shift gm and gain; re-center C_c/R_z on measured g_{m1}, g_{m6}.

Deliberate Divergences from Paper_1

Aspect	Paper_1	This design	Why
Inversion	Subthreshold	Strong/moderate	Project constraint
Rail	~1 V	1.8 V	Headroom for V_{ov}
Amp current	11.8 nA	4 µA	Strong inversion needs µA
Power	36 nW (whole PMU)	7.2 µW (amp)	Not a subthreshold target
Gain	46.4 dB (cascoded)	≥60 dB (simple 2-stage)	Better regulation, saves headroom
Sizing eqs	Subthreshold	Square law + Early-V	Different region
Reused	V_{th}, μC_{ox}, architecture, $V_{REF}=½V_{OUT}$, pass-PMOS, C_L concept		